

Sensor Mounting Methods

mmWave Presence Sensor Wall-Mount Reference

Project	Home Interior & Automation — Chitradurga
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Sensor	Sonoff SNZB-06PR2 (24GHz mmWave) or similar
Power	5V USB-C via Schneider USB module or HLK-PM01 buck
Stack	Sonoff ZBMINI R2 + Home Assistant + Zigbee2MQTT

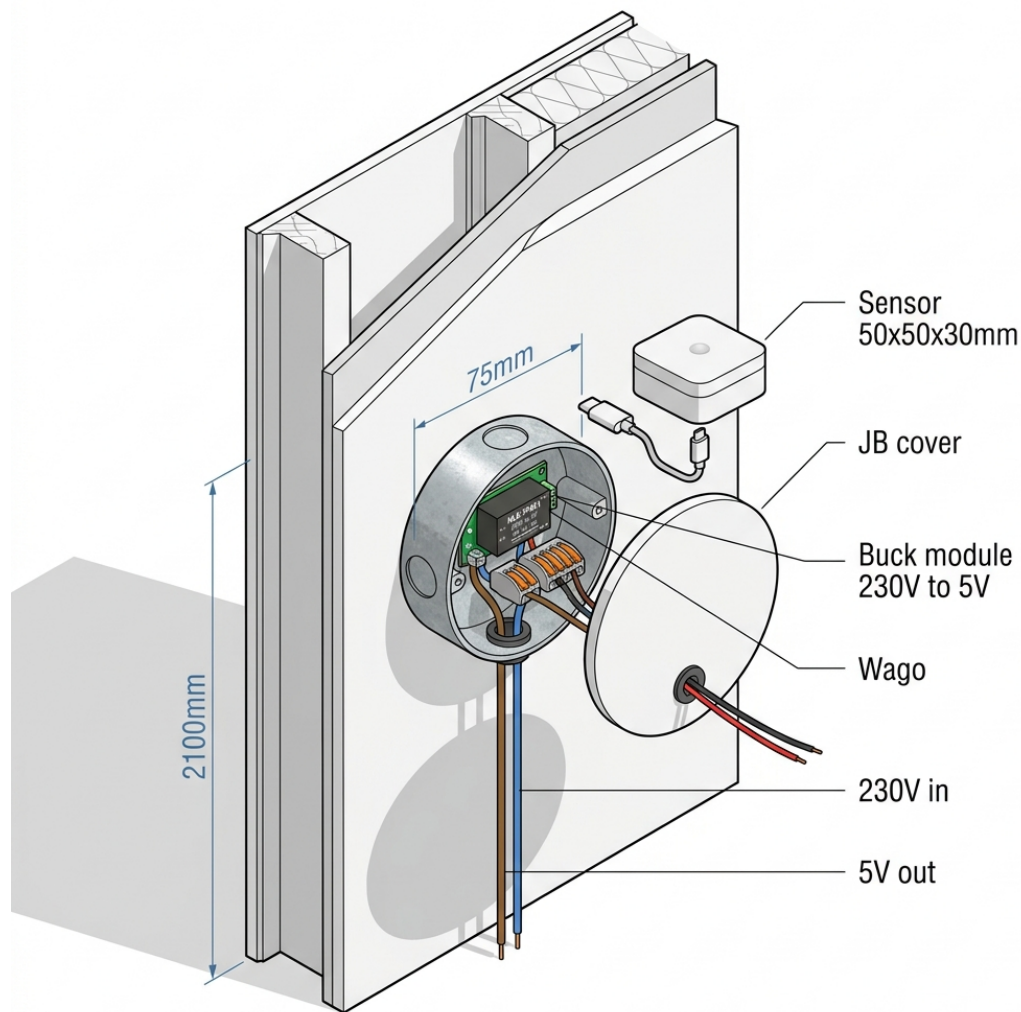
This document covers five approaches to mounting a wall-mounted mmWave presence sensor with a hidden USB-C power feed. Each method shows the components, layout, and trade-offs. The default recommendation for wall-mount installations is **Method B (bracket beside plate)**. For locations with an existing chase and junction box, **Method 1 (JB mount)** is the cleanest possible install.

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Method 1 — Junction Box Mount

BEST for GF Master Bathroom

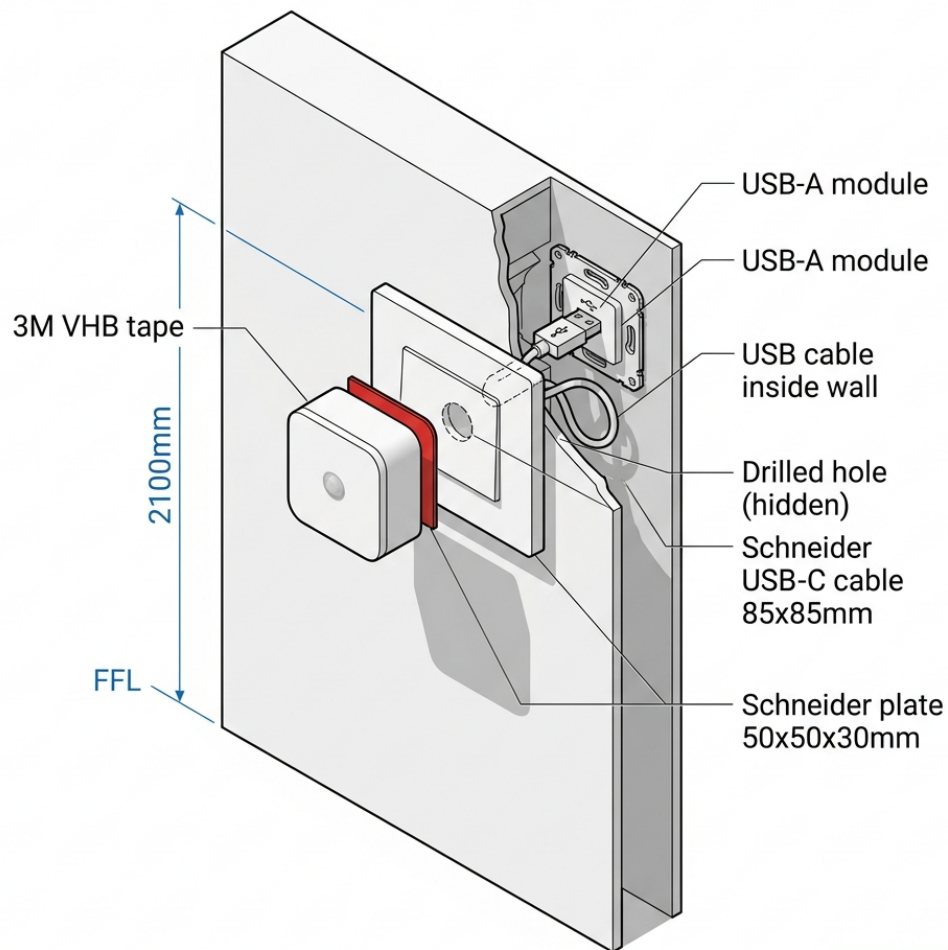


- Sensor mounts directly on a circular junction box at 2100mm FFL.
- HLK-PM01 buck converter (230V → 5V) lives **inside** the junction box.
- Zero visible cable — sensor sits flush on the JB cover.
- Wago 221 connectors for the 230V tap; 5V exits through a small grommet hole in the cover.
- **Best for:** locations with existing chase + JB provision (like GF master bathroom).

Method A — Direct Adhesive Mount

Sensor stuck to plate using 3M VHB tape

Front view

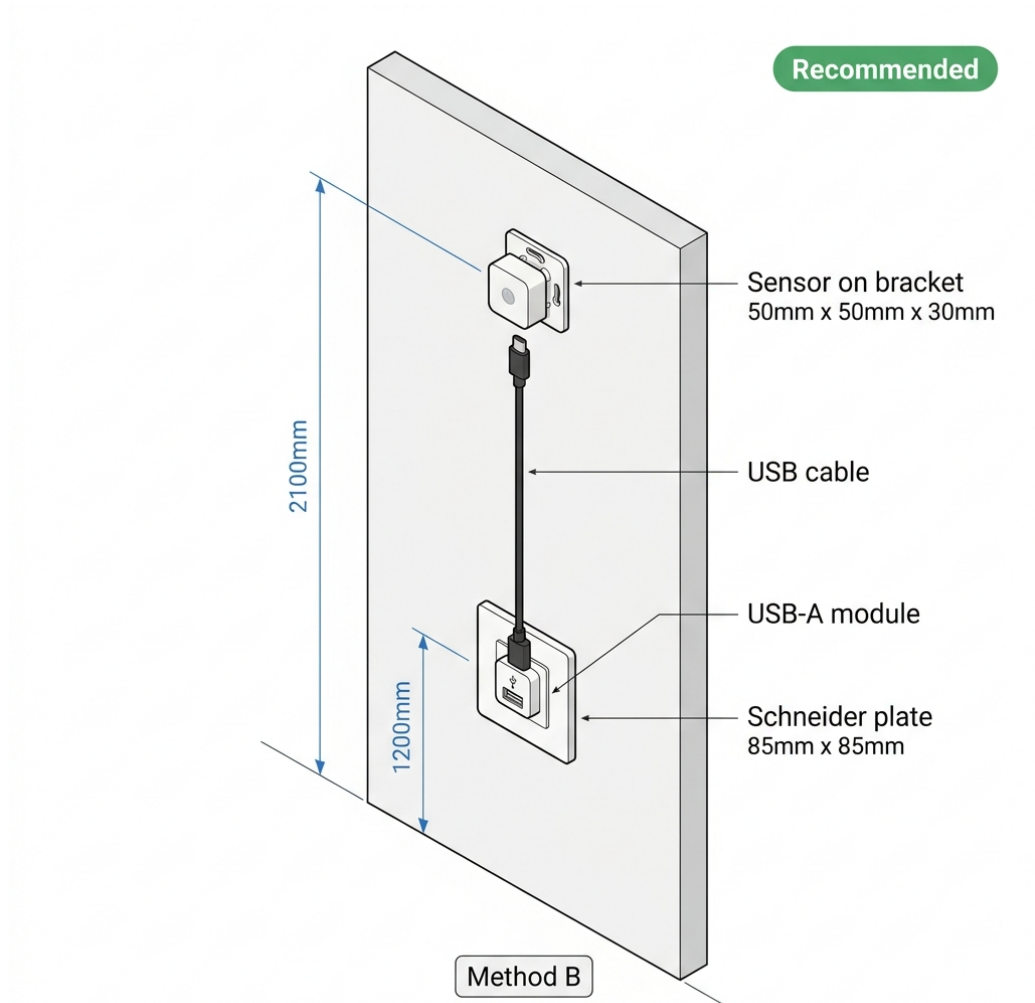


Method A

- Sensor stuck to the Schneider plate using 3M VHB double-sided tape.
- An 8mm hole is drilled through the plate (hidden behind the sensor body).
- USB-A to USB-C cable runs from the USB module inside the cavity, through the hole, into the sensor's back.
- Cheapest install but requires drilling the plate (non-reversible).
- **Trade-off:** sensor sits slightly proud on the plate; sensor's built-in magnetic mount is not used.

Method B — Bracket Beside Plate

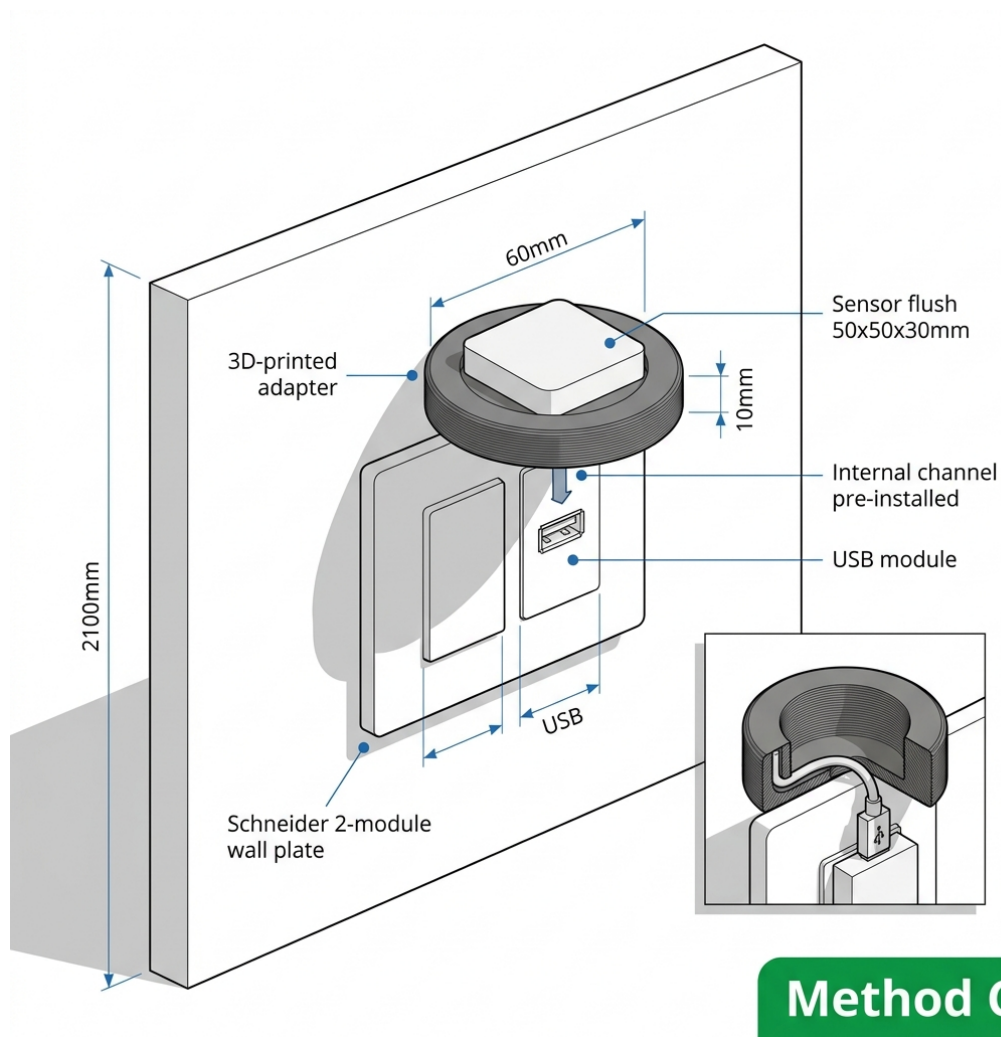
RECOMMENDED for most wall-mount sensor locations



- Sensor uses its **own supplied wall bracket** at 2100mm FFL.
- Schneider plate with USB-A module sits at 1200mm FFL.
- Short ~30cm USB-A to USB-C cable runs along the wall (tucked neatly).
- Looks deliberate and professional — like a designed install.
- No drilling of plates required, fully reversible.
- **Recommended** for FF bathrooms, bedrooms (wall option), and all future-provision rooms.

Method C — 3D-Printed Adapter

Custom adapter hides cable internally

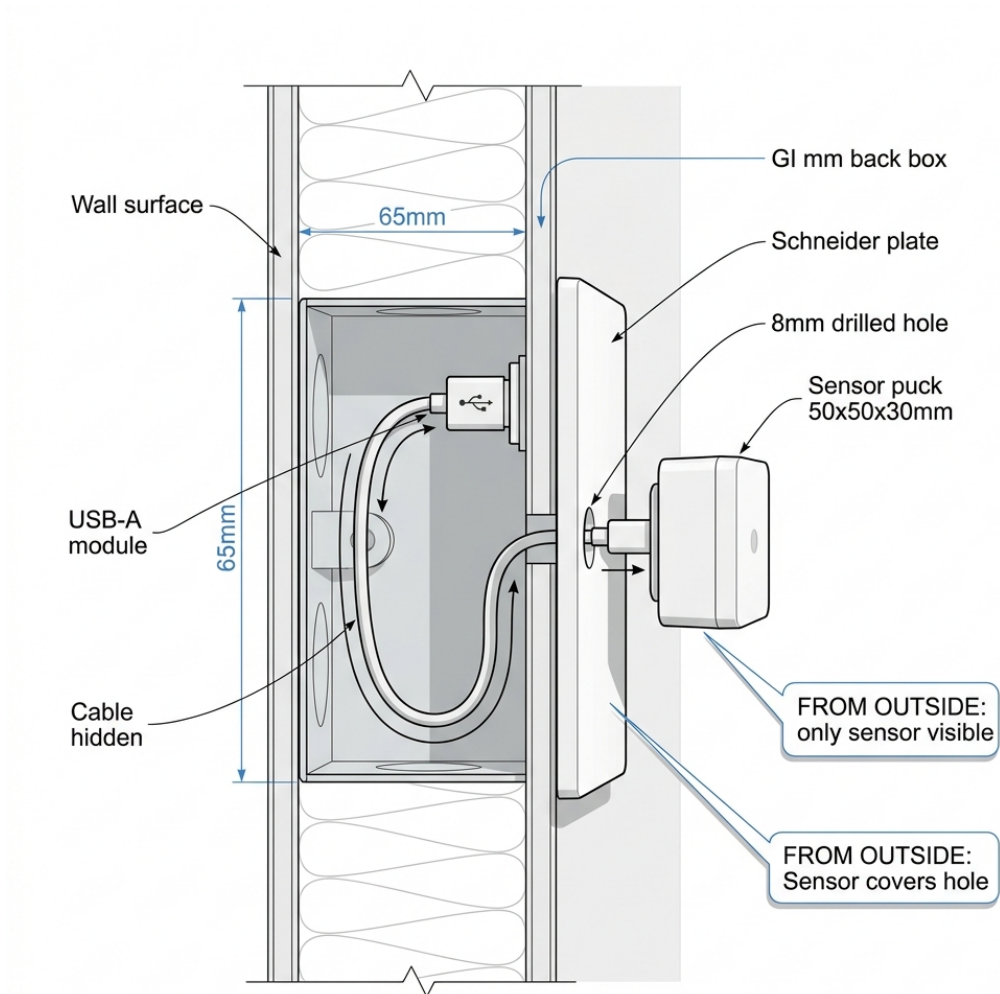


Method C

- Custom 3D-printed PLA adapter ring (60mm dia × 10mm thick) between plate and sensor.
- Internal cable channel hides the USB cable completely.
- Sensor sits flush on adapter — most polished look of all methods.
- Requires designing + printing the adapter (~\$200 online service print).
- **Best for:** spots where you want zero visible cable and have 3D printing access.

Cable Hiding Detail

Side-view cross-section showing cable path through plate



- Side view showing how the cable enters the wall cavity through the drilled hole.
- USB-A end plugs into the Schneider USB module inside the 65mm GI back box.
- USB-C end plugs into the sensor's back, through the drilled hole.
- Sensor body completely covers the hole — invisible from the front.
- Applies to Methods A and C (where the sensor sits ON the plate face).

Recommendation Matrix

Which method to use where

Location	Recommended Method	Why
GF Master Bathroom (existing JB)	Method 1 (JB Mount)	Cleanest install — leverages existing chase + JB infra; zero visible cable.
Bedrooms (false ceiling planned)	Ceiling-mount (separate doc)	Better coverage from ceiling; buck converter in switch cavity feeds via thin 2-core wire.
FF Bathrooms (concrete done)	Method B (Bracket beside plate)	No ceiling option available; bracket is fully reversible and looks deliberate.
Utility / Store (future provision)	Method B when installed later	Plan cavity + conduit + pull string now; install sensor when needed.
Dining / Living (big open space)	Ceiling-mount FP2 (separate doc)	Aqara FP2 multi-zone needs central ceiling location; powered via buck or PoE.

Notes for the electrician

- All sensor cavities should be marked at 2100mm FFL unless otherwise specified.
- Use 20mm PVC conduit for the sensor power run, with nylon pull-string left inside.
- Schneider USB-A module fits in any standard 1M slot of a 65mm GI MS back box.
- For JB Mount (Method 1): use a circular metal junction box; HLK-PM01 fits comfortably inside.
- All 230V terminations to use Wago 221 lever connectors (3-port for L branch).